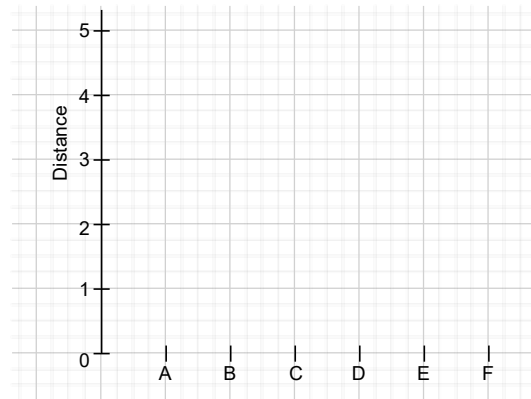
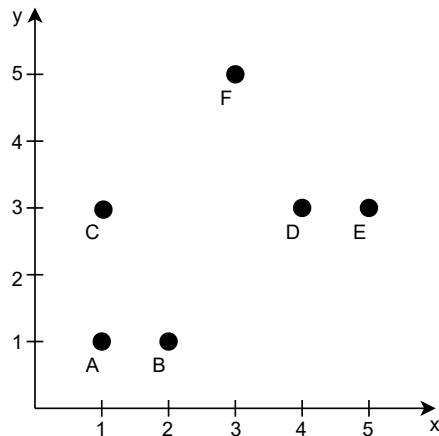
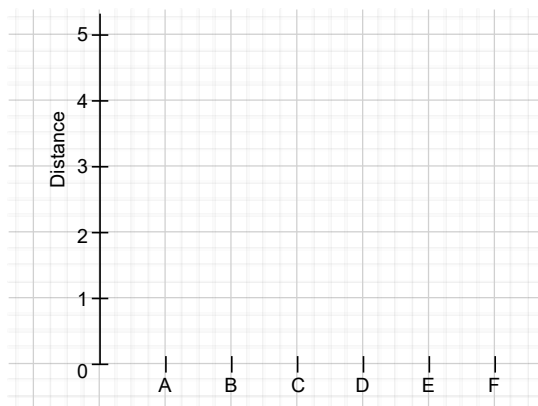


Exercise "Machine Learning - Clustering" of the lecture Digitization and Data Analytics

Task 1: Hierarchical Clustering

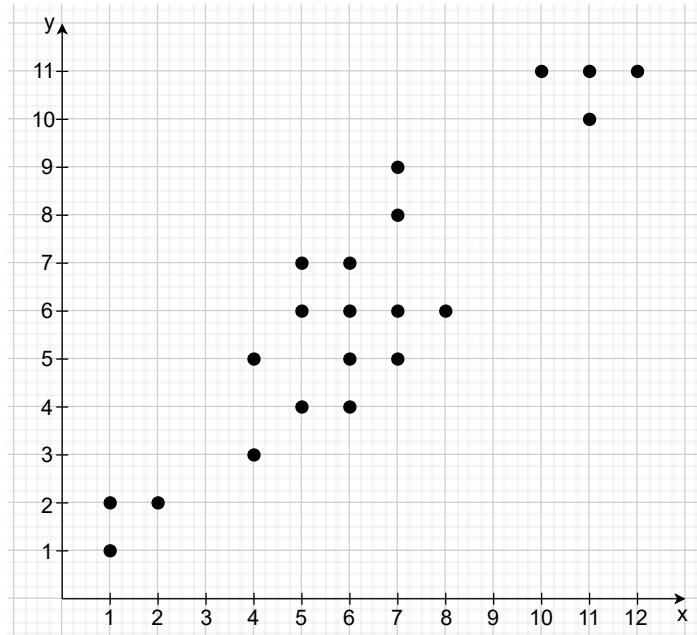


- a) You want to cluster the six points seen above. You decide that you want to create a dendrogram using Manhattan distance and average linkage. How does the resulting dendrogram look like? Complete the template given above!
- b) You want to cluster the same six points, but using maximum norm and complete linkage. How does the resulting dendrogram look like? Complete the template below!

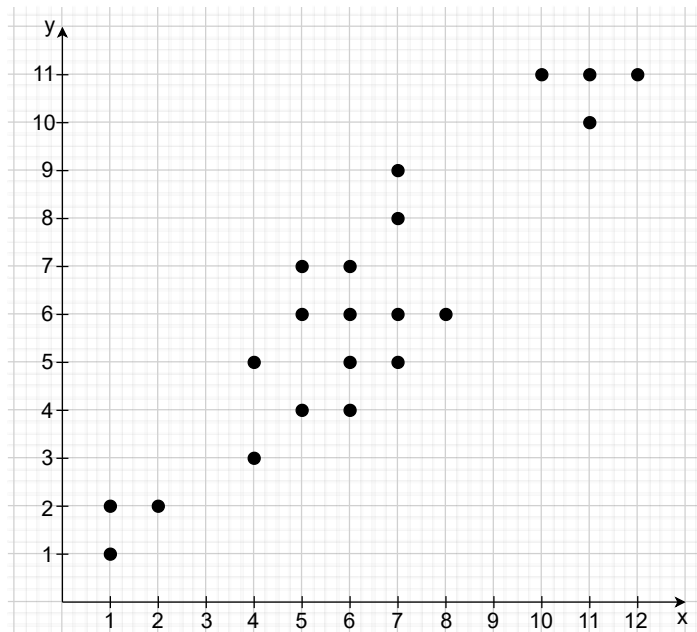


- c) You want two clusters to be formed with maximum norm and complete linkage. What distance do you have to choose? What points are in the clusters?

Task 2: Clustering with DBSCAN



- You want to cluster the points seen above. You decide that you want to use the density based clustering-algorithm DBSCAN. How would a clustering with $minPts = 4$ and $\epsilon = 2$ look like? Add the clusters to the Figure above. (Use euclidean distance.)
- You are not yet happy with the clustering result. So you try again with $minPts = 3$ and $\epsilon = 1$. How does the result look like? Add the clusters to the picture below.



- How do you have to choose $minPts$ and ϵ so that three clusters are formed and not a single point is classified as noise?